

HouseZero Solar & Battery Retrofit Optimisation

Hourly energy dispatch, retrofit economics and scenario optimisation in Excel

ANNUAL DEMAND

13,234 kWh

PV GENERATION

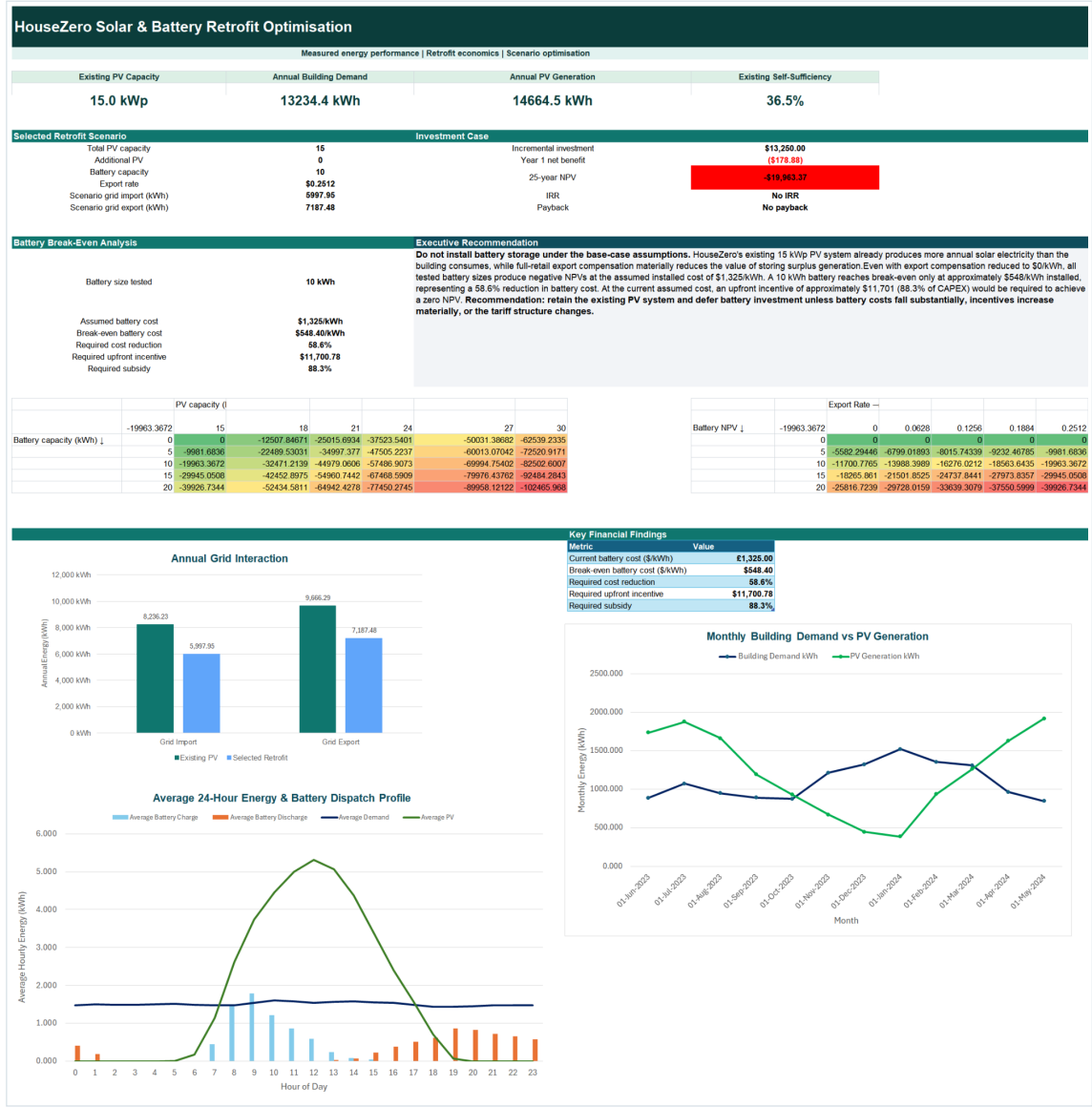
14,665 kWh

SELF-SUFFICIENCY

36.5%

25-YR NPV

-\$19,963



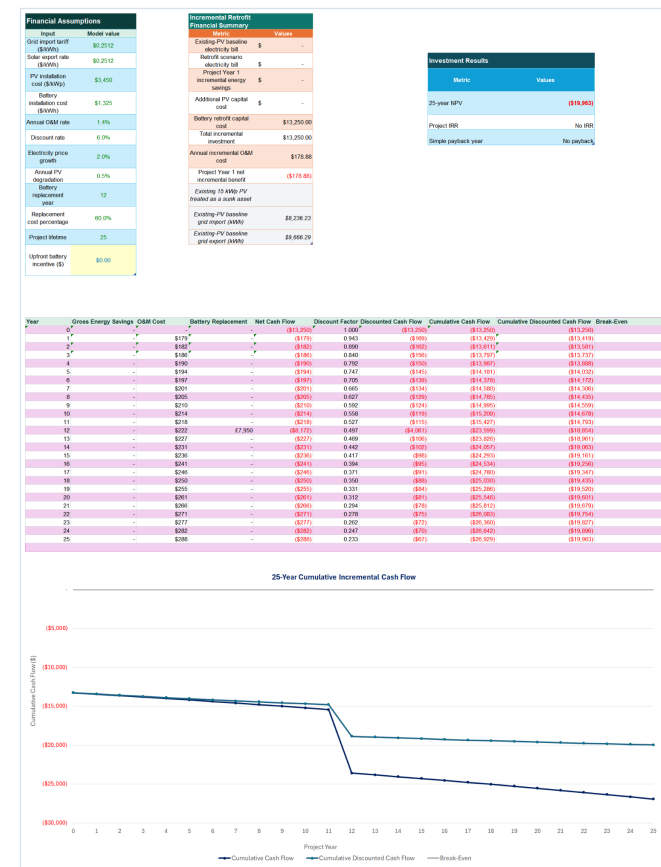
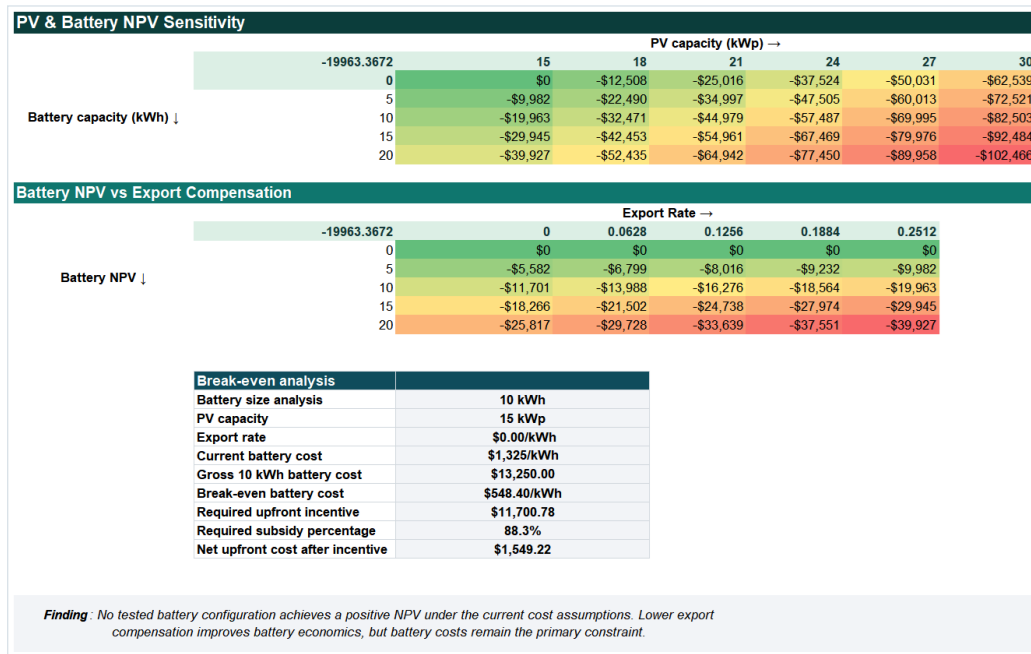
Base-case conclusion

Under the saved assumptions, adding a 10 kWh battery to the existing 15 kWp PV system is not financially attractive. The existing array already produces more annual solar electricity than the building consumes, and full-retail export compensation materially reduces the value of storing surplus generation.

Selected scenario

- 15 kWp PV + 10 kWh battery; incremental investment \$13,250.
- Grid import falls to about 5,998 kWh and grid export to about 7,187 kWh.
- Battery charges about 2,479 kWh and discharges about 2,238 kWh over the model year.
- The 25-year result is approximately -\$19,963 NPV with no IRR and no payback.

The model links physical dispatch constraints directly to 25-year incremental cash flows.



PV / battery and export-rate sensitivity

Incremental 25-year financial model

Methodology

- Power Query cleans the full-year hourly dataset (8,784 observations).
- Solar serves load first; surplus may charge the battery subject to power, efficiency and state-of-charge limits; residual surplus exports to the grid.
- The existing 15 kWp PV system is treated as a sunk asset. Incremental CAPEX includes only added PV and battery storage.
- Cash flows incorporate O&M, electricity-price growth, PV degradation, a year-12 battery replacement and a 6% discount rate over 25 years.
- Two-variable Excel Data Tables test PV/battery capacity combinations and export-compensation sensitivity.

Hourly calculation layer and audit trail

The raw dispatch layer remains visible so dashboard conclusions can be traced back to individual hours.

Hourly dispatch table

HouseZero Hourly Energy & Battery Dispatch Data																													
UTC_DateTime	Local_DateTime	Scenario_Start	Hour	Day_Type	Day_of_Week	Building_Demand_kWh	PV_Generation_kWh	Grid_Import_kWh	Grid_Export_kWh	PV_Meter1_kWh	PV_Meter2_kWh	Timestamp	Row	Scenario_PV_A	Street_PV_kWh	PV_Surplus_kWh	Local_Deficit_kWh	BODC_Start_kWh	Battery_Charge_kWh	Battery_Discharg_kWh	Battery_SOC_End	Scenario_Grid_In_kWh	Scenario_Grid_Export_kWh	Battery_Losses_kWh	SOC_Percent				
01-Jun-2023 04:00	01-Jun-2023 04:00	01-Jun-2023	0	Weekday	1	1274	0.000	1.274	0.000	1.274	0.000	45076.000	1	0.000	0.000	0.000	1.274	5.000	0.000	1.274	3.058	0.000	0.000	0.000	0.067	38.8%			
01-Jun-2023 05:00	01-Jun-2023 05:00	01-Jun-2023	1	Weekday	1	1454	0.000	1.454	0.000	1.454	0.000	45076.000	2	0.000	0.000	0.000	1.454	3.659	0.000	1.454	2.158	0.000	0.000	0.000	0.077	21.5%			
01-Jun-2023 06:00	01-Jun-2023 06:00	01-Jun-2023	2	Weekday	1	1448	0.000	1.448	0.000	1.448	0.000	45076.000	3	0.000	0.000	0.000	1.448	2.158	0.000	1.075	1.000	0.000	0.000	0.000	0.068	10.0%			
01-Jun-2023 07:00	01-Jun-2023 07:00	01-Jun-2023	3	Weekday	1	1398	0.000	1.398	0.000	1.398	0.000	45076.000	4	0.000	0.000	0.000	1.398	1.000	0.000	1.000	1.000	0.000	1.398	0.000	0.000	10.0%			
01-Jun-2023 08:00	01-Jun-2023 08:00	01-Jun-2023	4	Weekday	1	1616	0.000	1.616	0.000	1.616	0.000	45076.000	5	0.000	0.000	0.000	1.616	1.000	0.000	0.000	1.000	0.000	1.616	0.000	0.000	10.0%			
01-Jun-2023 09:00	01-Jun-2023 09:00	01-Jun-2023	5	Weekday	1	1397	0.004	1.313	0.000	1.313	0.048	45076.000	6	0.004	0.004	0.000	1.313	1.000	0.000	0.000	1.000	0.000	1.313	0.000	0.000	10.0%			
01-Jun-2023 10:00	01-Jun-2023 10:00	01-Jun-2023	6	Weekday	1	1398	0.000	1.398	0.000	1.398	0.000	45076.000	7	0.000	0.000	0.000	1.398	1.000	0.000	0.000	1.000	0.000	1.398	0.000	0.000	10.0%			
01-Jun-2023 11:00	01-Jun-2023 11:00	01-Jun-2023	7	Weekday	1	1400	0.000	1.400	0.000	1.400	0.000	45076.000	8	0.000	0.000	0.000	1.400	1.012	0.000	2.722	0.000	0.000	0.000	0.000	0.091	27.2%			
01-Jun-2023 12:00	01-Jun-2023 12:00	01-Jun-2023	8	Weekday	1	1349	5.782	0.000	4.413	4.413	3.183	45076.000	9	5.782	5.782	0.000	2.722	4.413	0.000	0.000	0.000	0.000	4.413	0.000	0.221	69.1%			
01-Jun-2023 13:00	01-Jun-2023 13:00	01-Jun-2023	9	Weekday	1	1004	7.158	0.000	5.354	5.444	3.800	45076.000	10	7.158	7.158	0.000	5.354	0.000	0.000	0.000	0.000	0.000	5.354	0.000	0.110	90.0%			
01-Jun-2023 14:00	01-Jun-2023 14:00	01-Jun-2023	10	Weekday	1	335	8.499	0.000	7.154	4.528	3.971	45076.000	11	8.499	8.499	0.000	3.95	7.154	0.000	0.000	0.000	0.000	7.154	0.000	0.000	90.0%			
01-Jun-2023 15:00	01-Jun-2023 15:00	01-Jun-2023	11	Weekday	1	1737	8.499	0.000	8.731	8.731	5.073	45076.000	12	8.499	8.499	0.000	5.073	8.731	0.000	0.000	0.000	0.000	8.731	0.000	0.000	90.0%			
01-Jun-2023 16:00	01-Jun-2023 16:00	01-Jun-2023	12	Weekday	1	1247	10.541	0.000	8.984	-8.984	5.195	45076.000	13	10.541	10.541	0.000	5.195	8.984	0.000	0.000	0.000	0.000	8.984	0.000	0.000	90.0%			
01-Jun-2023 17:00	01-Jun-2023 17:00	01-Jun-2023	13	Weekday	1	1420	10.027	0.000	8.608	-8.608	4.964	45076.000	14	10.027	10.027	0.000	4.964	8.608	0.000	0.000	0.000	0.000	8.608	0.000	0.000	90.0%			
01-Jun-2023 18:00	01-Jun-2023 18:00	01-Jun-2023	14	Weekday	1	1481	9.861	0.000	7.728	-7.728	4.481	45076.000	15	9.861	9.861	0.000	4.481	7.728	0.000	0.000	0.000	0.000	7.728	0.000	0.000	90.0%			
01-Jun-2023 19:00	01-Jun-2023 19:00	01-Jun-2023	15	Weekday	1	1390	7.624	0.000	6.732	-4.232	3.750	45076.000	16	7.624	7.624	0.000	3.750	6.732	0.000	0.000	0.000	0.000	6.732	0.000	0.000	90.0%			
01-Jun-2023 20:00	01-Jun-2023 20:00	01-Jun-2023	16	Weekday	1	228	6.899	0.000	6.823	-4.823	3.015	45076.000	17	6.899	6.899	0.000	3.015	6.899	0.000	0.000	0.000	0.000	6.899	0.000	0.000	90.0%			
01-Jun-2023 21:00	01-Jun-2023 21:00	01-Jun-2023	17	Weekday	1	1311	4.931	0.075	3.695	-3.020	2.448	45076.000	18	4.931	4.931	0.075	3.695	3.020	0.000	0.000	0.000	0.000	3.020	0.000	0.000	90.0%			
01-Jun-2023 22:00	01-Jun-2023 22:00	01-Jun-2023	18	Weekday	1	1004	2.922	0.000	1.918	-1.918	1.345	45076.000	19	2.922	2.922	0.000	1.345	1.918	0.000	0.000	0.000	0.000	1.918	0.000	0.000	90.0%			
01-Jun-2023 23:00	01-Jun-2023 23:00	01-Jun-2023	19	Weekday	1	1029	0.372	0.862	0.000	0.864	0.120	45076.000	20	0.372	0.372	0.862	0.000	0.864	0.000	0.000	0.000	0.000	0.864	0.000	0.000	90.0%			
02-Jun-2023 00:00	01-Jun-2023 20:00	01-Jun-2023	20	Weekday	1	1086	0.000	1.086	0.000	1.086	0.000	45076.000	21	0.000	0.000	0.000	1.086	0.000	0.000	0.000	0.000	0.000	1.086	0.000	0.000	90.0%			
02-Jun-2023 01:00	01-Jun-2023 21:00	01-Jun-2023	21	Weekday	1	1006	0.000	1.006	0.000	1.006	0.000	45076.000	22	0.000	0.000	0.000	1.006	0.000	0.000	0.000	0.000	0.000	1.006	0.000	0.000	90.0%			
02-Jun-2023 02:00	01-Jun-2023 22:00	01-Jun-2023	22	Weekday	1	1086	0.000	1.086	0.000	1.086	0.000	45076.000	23	0.000	0.000	0.000	1.086	0.000	0.000	0.000	0.000	0.000	1.086	0.000	0.000	90.0%			
02-Jun-2023 03:00	01-Jun-2023 23:00	01-Jun-2023	23	Weekday	1	1289	0.000	1.289	0.000	1.289	0.000	45076.000	24	0.000	0.000	0.000	1.289	0.000	0.000	0.000	0.000	0.000	1.289	0.000	0.000	90.0%			
02-Jun-2023 04:00	02-Jun-2023 00:00	02-Jun-2023	0	Weekday	1	1055	0.000	1.055	0.000	1.055	0.000	45076.000	25	0.000	0.000	0.000	1.055	0.000	0.000	0.000	0.000	0.000	1.055	0.000	0.000	90.0%			
02-Jun-2023 05:00	02-Jun-2023 01:00	02-Jun-2023	1	Weekday	1	1091	0.000	1.091	0.000	1.091	0.000	45076.000	26	0.000	0.000	0.000	1.091	0.000	0.000	0.000	0.000	0.000	1.091	0.000	0.000	90.0%			
02-Jun-2023 06:00	02-Jun-2023 02:00	02-Jun-2023	2	Weekday	1	1033	0.000	1.033	0.000	1.033	0.000	45076.000	27	0.000	0.000	0.000	1.033	0.000	0.000	0.000	0.000	0.000	1.033	0.000	0.000	90.0%			
02-Jun-2023 07:00	02-Jun-2023 03:00	02-Jun-2023	3	Weekday	1	1070	0.000	1.070	0.000	1.070	0.000	45076.000	28	0.000	0.000	0.000	1.070	0.000	0.000	0.000	0.000	0.000	1.070	0.000	0.000	90.0%			
02-Jun-2023 08:00	02-Jun-2023 04:00	02-Jun-2023	4	Weekday	1	1307	0.000	1.307	0.000	1.307	0.000	45076.000	29	0.000	0.000	0.000	1.307	0.000	0.000	0.000	0.000	0.000	1.307	0.000	0.000	90.0%			
02-Jun-2023 09:00	02-Jun-2023 05:00	02-Jun-2023	5	Weekday	1	1037	0.000	1.037	0.000	1.037	0.000	45076.000	30	0.000	0.000	0.000	1.037	0.000	0.000	0.000	0.000	0.000	1.037	0.000	0.000	90.0%			
02-Jun-2023 10:00	02-Jun-2023 06:00	02-Jun-2023	6	Weekday	1	1033	0.795	0.342	0.164	0.238	0.135	45076.000	31	0.795	0.795	0.342	0.164	0.238	0.000	0.000	0.000	0.000	0.238	0.000	0.000	10.0%			
02-Jun-2023 11:00	02-Jun-2023 07:00	02-Jun-2023	7	Weekday	1	1001	2.601	0.000	1.600	-1.600	1.275	45076.000	32	2.601	2.601	0.000	1.275	1.600	0.000	0.000	0.000	0.000	2.601	0.000	0.000	25.2%			
02-Jun-2023 12:00	02-Jun-2023 08:00	02-Jun-2023	8	Weekday	1	1122	4.739	0.000	3.617	-3.617	2.553	45076.000	33	4.739	4.739	0.000	2.553	3.617	0.000	0.000	0.000	0.000	3.617	0.000	0.000	39.8%			
02-Jun-2023 13:00	02-Jun-2023 09:00	02-Jun-2023	9	Weekday	1	1333	6.864	0.000	5.531	-5.531	3.825	45076.000	34	6.864	6.864	0.000	3.825	5.531	0.000	0.000	0.000	0.000	5.531	0.000	0.000	90.0%			
02-Jun-2023 14:00	02-Jun-2023 10:00	02-Jun-2023	10	Weekday	1	1113	8.138	0.000	7.017	-7.017	4.233	45076.000	35	8.138	8.138	0.000	4.233	7.017	0.000	0.000	0.000	0.000	7.017	0.000	0.000	90.0%			
02-Jun-2023 15:00	02-Jun-2023 11:00	02-Jun-2023	11	Weekday	1	1250	9.113	0.000	7.879	-7.879	4.701	45076.000	36	9.113	9.113	0.000	4.701	7.879	0.000	0.000	0.000	0.000	7.879	0.000	0.000	90.0%			
02-Jun-2023 16:00	02-Jun-2023 12:00	02-Jun-2023	12	Weekday	1	1322	9.687	0.000	8.364	-8.364	4.944	45076.000	37	9.687	9.687	0.000	4.944	8.364	0.000	0.000	0.000	0.000	8.364	0.000	0.000	90.0%			
02-Jun-2023 17:00	02-Jun-2023 13:00	02-Jun-2023	13	Weekday	1	1308	9.504	0.000	8.197	-8.197	4.712	45076.000	38	9.504	9.504	0.000	4.712	8.197	0.000	0.000	0.000	0.000	8.197	0.000	0.000	90.0%			
02-Jun-2023 18:00	02-Jun-2023 14:00	02-Jun-2023	14	Weekday	1	1222	8.797	0.000	7.575	-7.575	4.338	45076.000	39	8.797	8.797	0.000	4.338	7.575	0.000	0.000	0.000	0.000	7.575	0.000	0.000	90.0%			
02-Jun-2023 19:00	02-Jun-2023 15:00	02-Jun-2023	15	Weekday	1	1323	7.148	0.000	5.795	-5.795	3.510	45076.000	40	7.148	7.148	0.000	3.510	5.795	0.000	0.000	0.000	0.000	5.795	0.000	0.000	90.0%			
02-Jun-2023 20:00	02-Jun-2023 16:00	02-Jun-2023	16	Weekday	1	1800	2.482	0.014	1.435	-1.435	1.241	45076.000	41	2.482	2.482	0.014	1.435</												